R V COLLEGE OF ENGINEERING
(An autonomous institution affiliated to VTU, Belgaum)
DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

IM233AI – WORK SYSTEMS DESIGN
UNIT III
Topic ID 6: Work Measurement and Time Study

Q No	Question	Marks	Difficulty Level	BT Level	COs
1.	**Define a "Standard Time" according to Groover and explain the four specific conditions that must be met for a time standard to be valid.**	8	L	2	3
2.	**Rank the five primary work measurement methods by accuracy and justify the high ranking of Predetermined Motion Time Systems (PMTS).**	8	M	2	3
3.	**Rank the work measurement methods by "Relative Application Speed" (Time to set the standard) and explain why Historical Data is the fastest.**	8	M	2	3

Topic ID 7: Allowances and Time Study

4	**Distinguish between Personal, Fatigue, and Delay (PFD) allowances. Why are they necessary in determining the final standard time?**	8	M	2	1
5	**Explain the difference between "Constant Fatigue Allowances" and "Variable Fatigue Allowances." Provide examples of factors that influence variable fatigue.**	8	M	3	1
6	**What are "Contingency Allowances"? How do they differ from "Policy Allowances"?**	8	M	2	1
	A worker's observed time is 2.0 min, and their performance rating is 110%. The company uses a PFD allowance of 15%. Calculate the Normal Time and the Standard Time.	8	M	3	1
7	**Explain the "Internal" vs. "External" work elements in a worker-machine system. How do allowances apply differently to these?**	8	M	2	1
8	**Outline the step-by-step procedure for determining the Standard Time of a manual task starting from the initial observation.**	8	M	2	3
9	**Why is it necessary to break a task into "Work Elements" before timing it? Provide ANY six rules for defining these elements.**	8	M	2	3

<table>
<tr><td>10</td><td>An operator takes an average of 1.20 minutes to complete a cycle. The analyst rates the worker at 115%. If one element in the cycle was a machine-controlled process taking 0.40 minutes, calculate the Normal Time for the manual portion and the total cycle.</td><td>8</td><td>M</td><td>3</td><td>3</td></tr>
<tr><td>11</td><td>Discuss the procedure for determining standard time in "Machine-Paced" tasks. How do allowances differ for the machine portion vs. the manual portion?</td><td>8</td><td>M</td><td>2</td><td>3</td></tr>
<tr><td>12</td><td>The snap back timing method was used to obtain average times for work elements in one work cycle. The times are given below. Element d is a machine-controlled element and the time is constant. Elements a,b,c,e and f are operator controlled and were performance rated at 80%; however elements e and f are performed during the machine controlled element d.The machine allowance is zero and the PFD allowance is 14%. Determine the normal time and standard time for the cycle.
<table>
<tr><td>Element</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr>
<tr><td>Observed Time (min)</td><td>0.24</td><td>0.30</td><td>0.17</td><td>0.76</td><td>0.26</td><td>0.14</td></tr>
</table></td><td>8</td><td>M</td><td>3</td><td>3</td></tr>
<tr><td>13</td><td>The continuous timing method in direct time study was used to obtain the elemental times for a worker machine task as indicated.Element c is a machine controlled and the time is constant.Elements a,b ,d,e and f are operator controlled and external to machine cycle and they are performance rated at 80%.If the machine allowance is 25% and worker PFD allowance is 15%, determine the normal time and standard time for the work cycle.
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<tr><td>Observed Time(min)</td><td>0.18</td><td>0.30</td><td>0.88</td><td>1.12</td><td>1.55</td><td>1.80</td></tr>
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<tr><td>14</td><td>The continuous timing method in direct time study was used to obtain the elemental times for a worker machine task as indicated. Element c is a machine controlled and the time is constant. Elements a,b ,d,e and f are operator controlled and external to machine cycle and they are performance rated at 90%.If the machine allowance is 12% and worker PFD allowance is 8%, determine the normal time and standard time for the work cycle.
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<tr><td>Element</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr>
<tr><td>Observed Time(min)</td><td>0.24</td><td>0.38</td><td>0.88</td><td>1.04</td><td>1.65</td><td>1.90</td></tr>
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<tr><td>15</td><td>The continuous timing method in direct time study was used to obtain the elemental times for a worker machine task as indicated. Element c is a machine controlled and the time is constant. Elements a,b ,d,e and f are operator controlled and external to machine cycle and they are performance rated at 80%.If the machine allowance is 25% and worker PFD allowance is 15%, determine the normal time and standard time for the work cycle.
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Topic ID 8: PMTS

20	**Discuss the advantages of PMTS over direct stopwatch time study as highlighted by Groover.**	8	L	2	3
21	**Distinguish between first-level (MTM-1) and higher-level (MTM-2, MTM-3) PMT systems.**	8	M	2	3
22	**What is MOST (Maynard Operation Sequence Technique)? Explain its focus on "Sequence Models" rather than individual motions.**	8	M	2	3
23	**Break down the "General Move" sequence model (ABGABPA). Define each parameter.**	8	M	2	3
24	**Compare "General Move" vs. "Controlled Move" in Basic MOST. Give an example for each.**	8	M	3	3
25	**Distinguish between Mini-MOST, Basic MOST, and Maxi-MOST based on cycle length and repetitiveness.**	8	M	2	3
26	**What is "Admin MOST"? Provide three examples of tasks where it would be applied.**	8	M	3	3
27	**Discuss "MOST for Windows" (Computerized MOST). What are the advantages of using software over manual data cards?**	8	M	2	3
28	**Outline the systematic procedure for applying a Predetermined Motion Time System (PMTS) to establish a time standard for a manual task**.	8	L	2	3
29	**Describe the "Tool Use" sequence model. List four common tool-use actions it covers.**	8	M	2	3